
PO11Q - Introduction to Quantitative Political Analysis I
Exam 2025/26

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Exam Rubric
Time Allowed: 2 Hours
Exam Type: Written - Standard
Approved Calculators: Permitted

Answer all questions in the answer booklet provided

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Instructions

- Part **I** contains the questions for the exam
 - These are divided into four blocks: A, B, C, and D.
 - Block A is compulsory and must be answered by all candidates.
 - You must then select ONE block out of B, C, and D.
 - Should you change your mind during the exam, indicate clearly which block you wish to have marked.
 - You can achieve a maximum of 120 points.
 - Information on the maximum number of points obtainable is outlined at the beginning of each block
- Part **II** contains the case study to which all questions in Block A of Part **I** refer
- Part **III** contains the relevant statistical tables
 - Normal curve tail probabilities are provided in Table **4**, page **11**.
 - t-distribution critical values are provided in Table **5**, page **12**.
- Part **IV** contains the formula collection and notation of the module
 - If you use notation other than in the formula collection (page **13**) or notation table (page **13**), give an explanation.

Part I: Exam Questions

1 | Block A – Compulsory

80 points, points for each question according to level

All questions in this section relate to the case study in Part II.

Level 1 (each question 4 points)

1. How many countries are in the `ssa` data frame? How do you know?
2. Section 9: Which shape does the distribution of `enrol_net` have? Explain your answer.

Level 2 (each question 6 points)

3. What was the probability of a country to be democratic in sub-Saharan Africa in 2000? How do you know?
4. How is the following code chunk different from that in Subsection 8.1 and what result would you obtain from the one presented here? Explain your answer.

```
ssa <- filter(wdi, un_region_name == "Southern Africa" &  
              un_region_name == "Middle Africa" &  
              un_region_name == "Western Africa" &  
              un_region_name == "Eastern Africa")
```

5. Interpret the output in Section 13.
6. Verbalise the code in Subsection 11.1, taking care to explain the value range of "low" and "high".
7. Subsection 11.2: Name and explain the numerical value that could replace

```
quantile(ssa$gdppc, probs = 0.5)
```

8. Section 13: Are the results of the significance test statistically robust? Why / why not?
9. Subsection 8.2: State the result of

```
ssa_new$country[nrow(ssa_new)]
```

10. Why does the code in Subsection 11.3 use `labels=` and not `levels=`?

Block A is continued on page 3.

Level 3 (each question 8 points)

11. Consider Figure 2 in Subsection 12.2. How does the code underneath have to be adjusted in order to obtain the graph as depicted? You are welcome to use code in the answer, but it is sufficient to explain what needs changing in plain English.
12. Section 13: What is the standard deviation of `gdppc` in North Africa? Show your reasoning.
13. Draw a box-and-whisker plot for `life`.

Bonus (4 points, you can achieve a maximum of 120 points)

- Who does Flo's desktop background show?

2 | Block B – True / False Statements

40 points, 5 points for each statement

State whether each of the following statements is true or false. Explain why the statement is true or false.

1. The α -level balances the probability between committing a Type I and a Type II Error.
2. The test of a theory in empirical social science research relates to the mechanisms approach of causality.
3. A confidence interval and a significance test are essentially the same thing.
4. In causality, symmetry is the opposite of asymmetry.
5. The Central Limit Theorem postulates that the sampling distribution is always normal even if the population distribution is not normal.
6. Compared to inferential statistics, descriptive statistics are unimportant.
7. The null-hypothesis always indicates no effect.
8. The smaller the sample size, the greater the uncertainty in inference.

3 | Block C – Calculations

40 points, points for each question in parentheses

For the following questions you are required to carry out calculations. Show the calculations in detail and explain the steps you are taking.

1. Using the data presented in Table 1, calculate:
 - a. the mean of x . **(2)**
 - b. the median of x . **(2)**
 - c. the mode of x . **(2)**
 - d. the variance of x . **(5)**
 - e. the standard deviation of x . **(2)**
 - f. the 30th percentile of x . **(5)**
2. Assume that the values of x in Table 1 represent the total amount of campaign spending (in thousands of £) in each of the five electoral districts of a small country during the most recent parliamentary election. Assess how atypical the spending in District 5 was relative to the spending pattern across all districts. **(6)**
3. A political scientist surveys a sample of 25 municipal government employees to estimate the average number of policy reports they produce per year. The sample yields an average of 37 reports, with a standard deviation of 6 reports. What is the lower limit of the 95% confidence interval for the average number of reports produced annually by all municipal government employees? Round your answer to three decimal places. **(6)**
4. In the year 2000, the average per capita GDP of sub-Saharan Africa was 972.49 US\$ with a standard deviation of 1451.44. At a 95% confidence level, was this value statistically different from the average per capita GDP of North Africa which was 2402.232 US\$ for the same period? **(10)**

Obs.	x
1	141
2	460
3	183
4	160
5	736

Table 1: Data Set - Block C

4 | Block D – Explanations

40 points, 8 points for each method / concept

Explain each of the following methods or concepts in your own words, and illustrate your explanation with a worked example of your choice.

1. Sampling Distribution
2. $E[x]$
3. Confidence Interval
4. Counterfactual
5. Systematised Concept

Part II: Case Study

5 | Preface

The data set is composed of three different sources joined together:

- World Development Indicators: The World Development Indicators is a compilation of relevant, high-quality, and internationally comparable statistics about global development and the fight against poverty. The database contains 1,400 time series indicators for 217 economies and more than 40 country groups, with data for many indicators going back more than 50 years. (The World Bank, n.d.)
- BMR: This data set provides a dichotomous coding of democracy for every country in the world (222 total) from 1800 to 2020. (Miller et al., 2022)

The data in the data set is for the year 2000.

6 | Code Book

variable	label
country	Country Name
democracy	0 = Autocracy, 1 = Democracy (Miller et al., 2022)
gdppc	GDP per capita (current US Dollars)
enrol_net	School enrollment, primary (% net)
life	Life expectancy at birth, total (years)
un_region_name	Region on continent

Table 2: Variables in the WDI_PO11Q data set for 2000

7 | Preliminaries

```
wdi <- read.csv("world.csv")  
  
library(tidyverse)
```

8 | Data Manipulation

8.1 Sub-Setting

```
ssa <- filter(wdi, un_region_name == "Southern Africa" |  
              un_region_name == "Middle Africa" |  
              un_region_name == "Western Africa" |  
              un_region_name == "Eastern Africa")  
  
north_africa <- filter(wdi, un_region_name == "Northern Africa")
```

8.2 Data Arranging

```
ssa_new <- ssa %>%  
  arrange(desc(gdppc))
```

9 | Descriptives for ssa Data Frame

	N	Mean	SD	Min	P25	Median	P75	Max
gdppc	46	972.49	1451.44	122.27	263.61	465.10	730.35	8063.66
enrol_net	31	63.24	19.13	26.83	52.09	63.56	79.61	99.05
life	46	53.62	6.47	44.83	48.88	52.34	56.86	72.78
democracy	46	0.30	0.47	0.00	0.00	0.00	1.00	1.00

Table 3: Descriptive Statistics for 'ssa' Data Frame

10 | Proportion of missing values per variable

```
##      country      gdppc      enrol_net      life      democracy  
##      0.000000      0.000000      0.326087      0.000000      0.000000  
## un_region_name  
##      0.000000
```

11 | Recoding

11.1 Life expectancy at Birth

```
ssa <- ssa %>%  
  mutate(lifecat=  
    ordered(  
      cut(life, breaks=c(0,mean(ssa$life),80),  
        labels=c("low","high"))))
```

11.2 per capita GDP

```
ssa <- ssa %>%  
  mutate(gdpcat=  
    ordered(  
      cut(gdppc, breaks=c(0,quantile(ssa$gdppc, probs = 0.5),8064),  
        labels=c("low","high"))))
```

11.3 Democracy

```
ssa$democracy <- factor(ssa$democracy, labels=c("Autocracy", "Democracy"))
```

12 | Visualisations

12.1 Cumulative Distribution Function

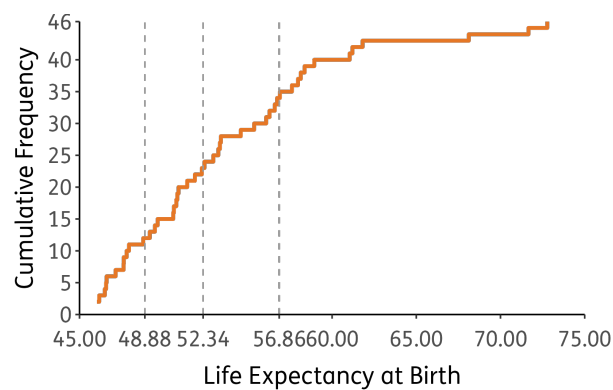


Figure 1: Cumulative Frequency of Life Expectancy in sub-Saharan Africa (2000)

12.2 Bar Graph

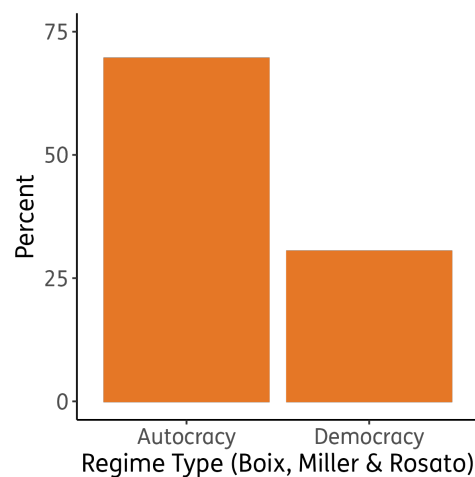


Figure 2: Regime Distribution in sub-Saharan Africa (2000)

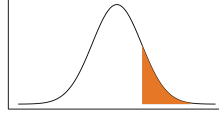
```
ggplot(data=ssa)+  
  geom_bar(aes(x=democracy), color="orange", fill="orange")+  
  scale_y_continuous(name="Percent",  
    limits = c(0,100),  
    breaks = c(0,25,50,75,100))
```

13 | Inference

```
t.test(north_africa$gdppc,  
mu = mean(ssa$gdppc, na.rm = TRUE))  
##  
## One Sample t-test  
##  
## data: north_africa$gdppc  
## t = 1.4383, df = 5, p-value = 0.2099  
## alternative hypothesis: true mean is not equal to 972.4918  
## 95 percent confidence interval:  
## -152.9952 4957.4588  
## sample estimates:  
## mean of x  
## 2402.232
```

Part III: Statistical Tables

14 | Normal-Distribution



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002

Table 4: Right-Tail Probabilities under the Normal Distribution

df	Confidence Level									
	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	One-Tail Probability									
	$t_{0.25}$	$t_{0.20}$	$t_{0.15}$	$t_{0.10}$	$t_{0.05}$	$t_{0.025}$	$t_{0.01}$	$t_{0.005}$	$t_{0.001}$	$t_{0.0005}$
1	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.22	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Table 5: Probabilities under the t-Distribution

Part IV: Formulae and Notation

16 | Formulae

Statistic	Formula
Confidence Interval	$Pr(\bar{y} - t_{\alpha/2} \cdot se \leq \mu \leq \bar{y} + t_{\alpha/2} \cdot se) = 1 - \alpha$
Deviation	$d = y_i - \bar{y}$
Mean	$\bar{y} = \frac{\sum y_i}{n}$ $\mu = \sum yP(y) = E[y]$
Position of p th percentile	$P = (n + 1) \cdot \frac{p}{100}$
Range	$y_{range} = y_{max} - y_{min}$
Standard Deviation	$s = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n-1}}$
Standard Error	$\sigma_{\bar{y}} = \frac{\sigma}{\sqrt{n}}$ $se = \frac{s}{\sqrt{n}}$
t-test	$t = \frac{\bar{y} - \mu_0}{se}$
Variance	$s^2 = \frac{\sum (y_i - \bar{y})^2}{n-1}$
z-score	$z = \frac{y - \mu}{\sigma}$

Table 6: Formulae for PO11Q

17 | Notation

Symbol	Explanation
d	Deviation
n	Sample Size
s	Standard Deviation
s^2	Variance
$\bar{y}$	Mean
y_i	Observation i
f	(Absolute) Frequency
cf	Cumulative (Absolute) Frequency
rf	Relative Frequency
crf	Cumulative Relative Frequency
$E[x]$	The expected value of x
μ	Mean of the Population
se	Standard Error (with s of sample)
σ	Standard Deviation of the Population
$\sigma_{\bar{y}}$	Standard Error (with σ of population)
t	t-value
z	z-value

Table 7: Notation

Part V: References

List of References

- Miller, M., Boix, C., & Rosato, S. (2022). *Boix-Miller-Rosato Dichotomous Coding of Democracy, 1800-2020*.
<https://doi.org/10.7910/DVN/FENWWR>
- The World Bank. (n.d.). World Development Indicators. <https://datacatalog.worldbank.org/dataset/world-development-indicators>